MILITARY SPECIFCATION

FUZE, ROCKET, M423, MODIFIED (BUNKER) LESS BOOSTER AND LEAD

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the parts and assembly drawings of a point detonating fuze, modified for bunker penetration, designated as Fuze, Rocket, M423, Modified (Broker) Less Booster and Booster Lead (see 6.7).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys
MIL-D-18665 - Detonator, Mark 59

PURCHASE DESCRIPITION

AEI-93 - Primer, M104 (XM94E1) Parts for and Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331 - Fuze and Fuze Components, Environmental and Perforformance Tests for

FSC: 1340

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MIL-STD-810 - Environmental Test Methods for Aerospace and Ground Equipment.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

9271602 - Fuze, Pocket, M423, Modified (Bunker) Less Booster and Booster Lead
9275050 - Box, Fiberboard, Packing, Ammunition for Fuze, Rocket, M423 Modifid (Bunker)
9275051 - Box, Wirebound, Packing, Ammunition for Fuze, Rocket, M423 Modified (Bunker)

PUBLICATIONS

ORNANCE CORPS

ET - 9271602 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Conformance. - The fuze assemblies and parts shall comply with all requirements specified on Drawing (dwg.) 9271602, and with all requirements specified in applicable specifications.

3.2.1 Spring embrittlement .- None of the springs used in the fabrication of the fuze shall break when extended or distorted to the point of minimum permanent distortion.

3.2.2 Setback weight spring set. - The setback weight spring shall return within .015 inch of the original free length when compressed and heat treated.

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3.3 Safety and Arming Device

3. 3.1 Arming and Detent Functioning. - The rotor assembly in the loaded Safety and Arming Device (S&A) shall rotate from the fully safe position to the fully armed position and the detent shall lock the rotor in the fully armed position.

3.3.2 Setback weight return capability.- The setback weight in the loaded safety and arming device shall return to the fully safe position after being bottomed with a 20 pound force and released.

3.3.3 Rotor reset capability. - The rotor assembly in the loaded safety and arming device shall be capable of return to the fully safe position, after being rotated to a position of 26 degrees but not more than 35 degrees from the fully safe position and released.

3.3.4 Arming Time. - The rotor assembly in the loaded S&A shall rotate from the fully safe position to the fully armed position in not less than 0.67 sec. and not more than 0.81 sec. when subjected to arming acceleration of 27.00 plus or minus 0.25 g's.

3.3.5 Non-Arm. - The rotor assembly in the loaded S&A shall not arm when subject to acceleration of 11.25 g's minimum for a minimum of 3 seconds. This test shall be conducted just prior to insertion of the S&A into the fuze body.

3.3.6 S&A position. - The safe position of the S&A shall be verified after insertion into the fuze body assembly. After insertion of the S&A fuze handling shall be positively controlled to preclude any chance of inadvertant arming or substitution of the S&A until assembly of the booster housing.

3.4 Complete Fuze Assembly

3.4.1 Arming Time. - The fuze assembly shall arm in not less than 0.63 sec. or more than 0.82 sec. when subjected to arming acceleration of 27.00 plus or minus 0.25 g's and the detent shall lock the rotor in the fully armed position, as determined by radiographic inspection.

3.4.2 Non-Arming. - The fuze assembly shall not arm when subjected to an acceleration of 11.50 g's plus or minus 0.25 g's for a minimum of 3 seconds.

3.4.3 Jolt and Jumble. - The fuze assembly shall be subjected to and must withstand the Jolt test specified in MIL-STD-331, with

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the exception that the fuze shall not be disassembled and inspected after the Jolt test. After the completion of the Jolt test, the same fuze shall be subjected to and must withstand the Jumble test specified in MIL-STD-331.

3.4.4 Forty-foot drop. - The complete fuze assembly shall withstand the forty-foot drop test specified in MIL-STD-331. To satisfy the requirements of MIL-STD-331, the disassembled fuze must display the following conditions:

a. No loose explosive shall be present in the fuze body assembly.

b. The lock roller pin on the rotor shall not be broken.

c. The rotor shall be in the safe position, secured such that it cannot be turned to the armed position with finger pressure.

d. The g-weight shall be in fully safe position.

3.4.5 Operational Tests. - The unarmed fuzes shall be subjected to and must withstand the following operational tests: Five Foot Drop Test as specified in MIL-STD-331; Non-Environment Test - Non-Arm and Arming Times only: Functional Test - Aircraft Vibration, Method 514, Curve Z, as specified in MIL-STD-810. The vibration conditioning shall be continuously performed until completed. After completion of the Five Foot Drop Test and the Functional Test, including the acceptance criteria for these tests (leak test, non-arming test, and arming time test), the same fuzes shall be tested for high order functioning.

3.4.6 Seal leakage test. - The leakage in the fuze shall not cause a pressure drop greater than 0.00072 psi in 10 sec.

3.4.7 Fuze Firing

3.4.7.1 Point Detonating Mode. - The fuze assembly shall function the explosive elements when a weight is dropped onto the nose of the fuze.

3.4.7.2. Bunker Mode. - The fuze assembly in the bunker mode shall function the explosive elements when the fuze is dropped nose-down onto a steel plate.

3.5 Workmanship. - All parts and assemblies shall be free from burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, porosity, warpage, burn marks, checks, blisters, excess flash, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agents. Surface coating shall be continuous except for a few slight scratches not exposing base material. The loading and assembly operations shall be performed in a thorough workmanlike manner consistent with the best practices of the industry. All required marking shall be neat and sharply defined.

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3.6 First article inspection. - This specification makes provisions for first article inspection. Requirement for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PREVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Submission of product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract , this authority is vested in another agent of the certifying organization.

4.1.2 Contractor inspection system. - As part of the contractor's approved quality procedure, there shall be a provision for checking the adequacy of inspections made in connection with listed critical or major defects. The procedures shall include the following provisions:

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4.1.2.1 Use of simulated defect standards. - The following listed verified inspections shall be validated and inspection equipment verified hourly using Design Agency approved item facsimiles or standards containing the defects and supplied by the contractor. The contractor shall resupply the item facsimiles when it is determined by the Government that the facsimilies are no longer representative enough of the product to make it unidentifiable to the contractor inspection personnel, or it is not functioning in the necessary manner and cannot be corrected satisfactorily. These standards should be representative of defects actually found or anticipated. They should be inert and readily identifiable to the Government to ensure recovery should they be inadvertently introduced into production. Simulated defect standards used to verify the following verified inspections should be constructed and identified in such a manner that they are not readily recognizable to the contractor inspection personnel as defect standards. Standards should be used, so that their number, time, and sequence of introduction into the inspection cycle is known only to those personnel responsible for checking the accuracy of the operation. Adequacy of the standards shall be verified at least once each shift.

Verified Inspections

Paragraph 4.3.3.11 - Setback weight return capability.
Paragraph 4.3.3.15 - S&A position check.

NOTE: When there is automatic detection and automatic rejection of defective items to a controlled area or a controlled rejection of defective items, it is permissible for the defect standards to be identifiable and their introduction into production noticeable to contractor inspection personnel.

4.1.2.2 Undetected defect standards. Provisions shall be established for the corrective action to be taken where the equipment or contractor inspection personnel fail to identify a standard as defective. These provisions shall include a reinspection of all acceptable material subsequent to the last acceptable verification check as a minimum.

4.2 First Article Inspection

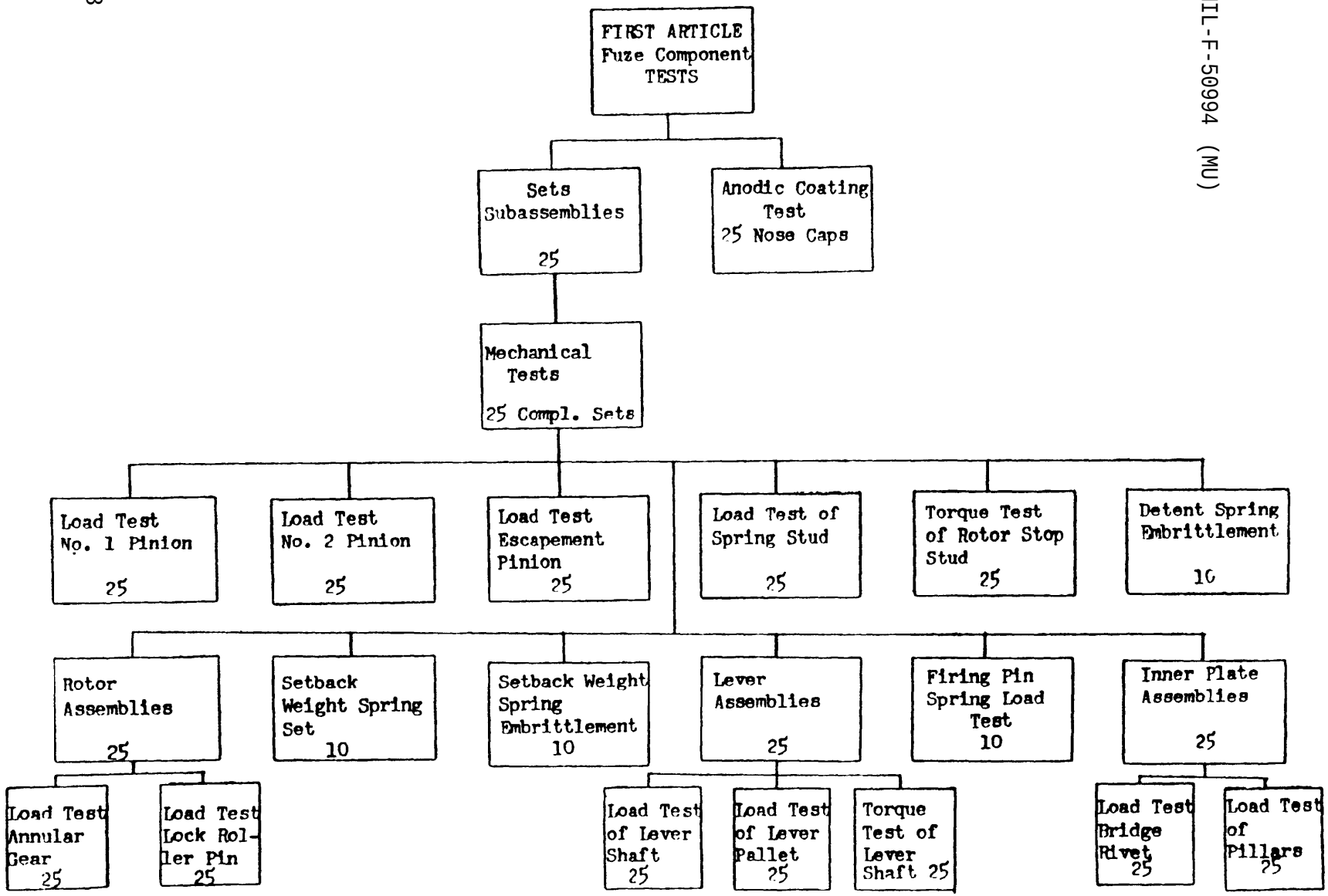
4.2.1 Submission. - The contractor shall submit a first article sample as designated by the contracting officer to a Government approved facility for evaluation in accordance with the provisions of 4.2.2. The first article sample shall have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production process, pro-

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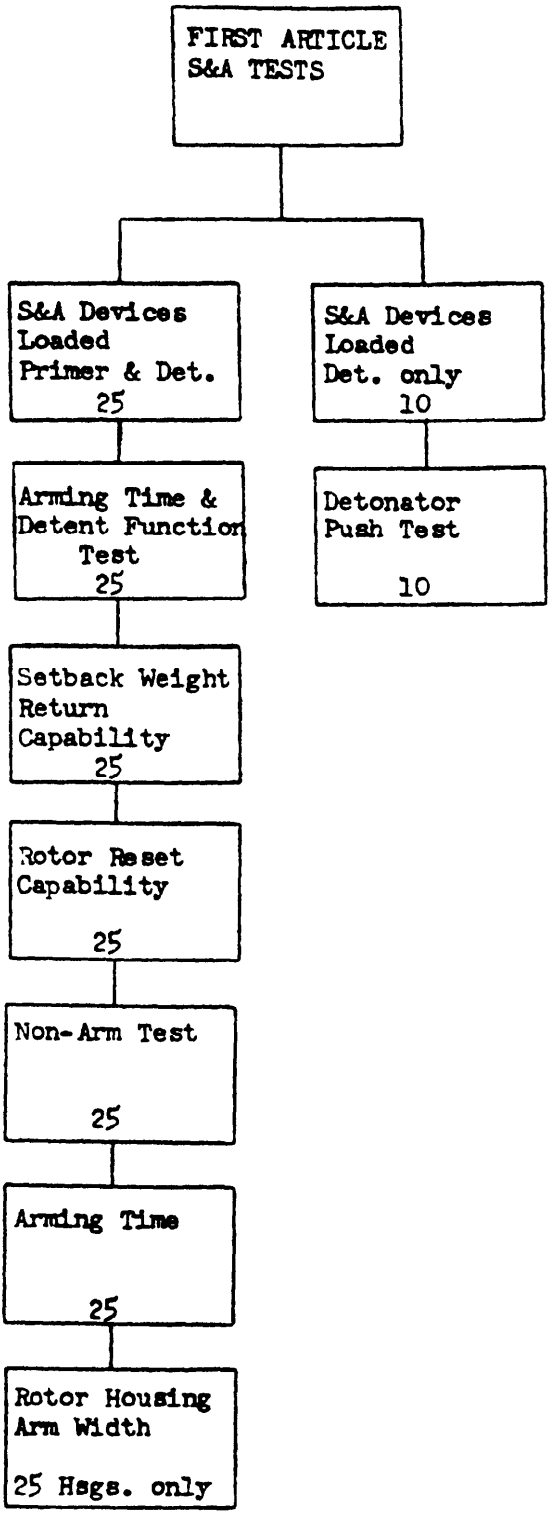
cedures, and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conform to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings/ certificates of conformance for materials (see 4.1.1.c). A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 Inspection to be performed. - Each fuze, subassembly, and component may be examined for every drawing dimension and drawing requirement applicable to the fuze, subassemblies, or components. In addition, any or all of the following examinations and tests will be performed as applicable.

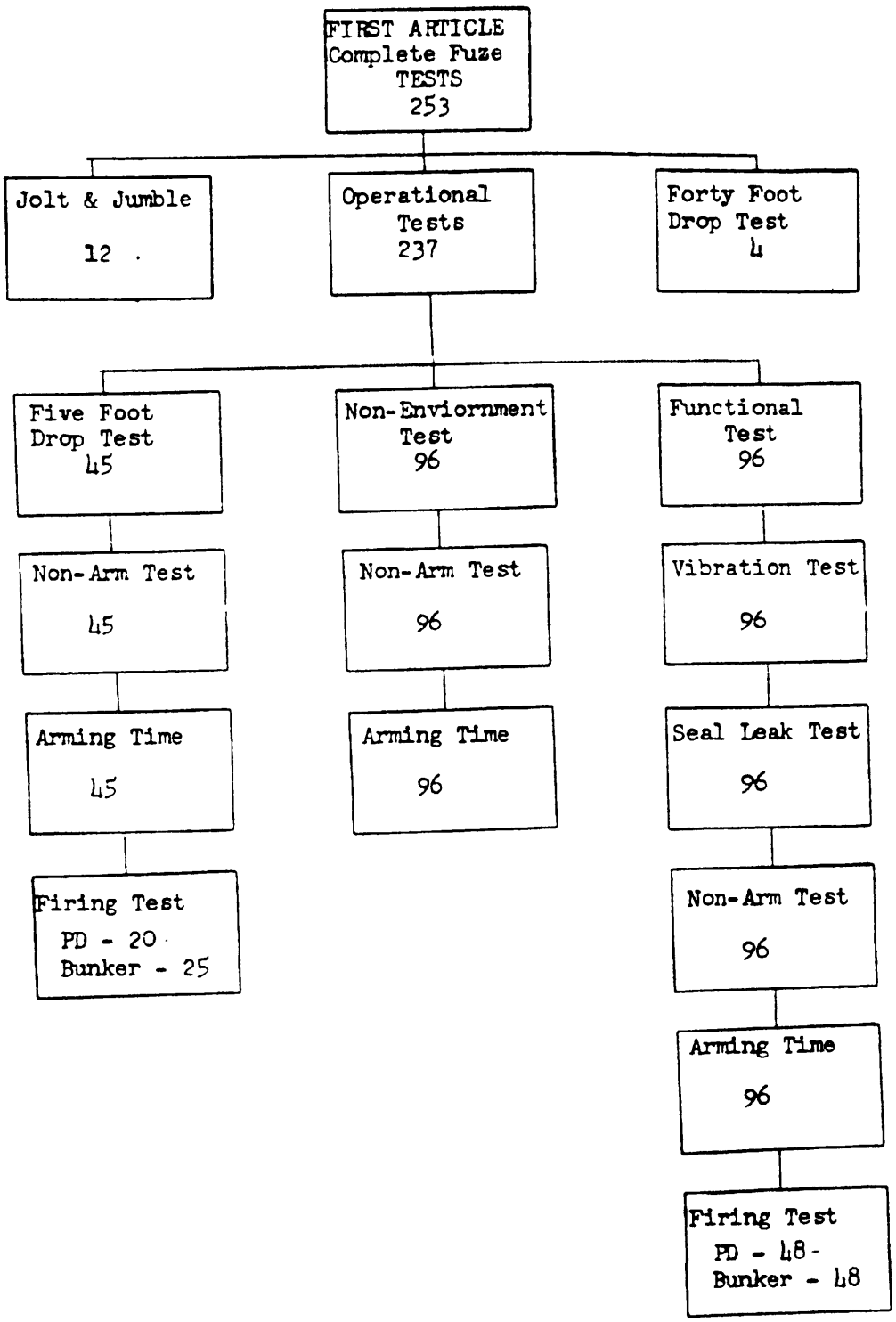
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4.2.3 Rejection of first article sample. - If any fuze, subassembly, or component fails to comply with the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a fuze, subassembly or component in the sample to comply with any of the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is in no way authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded fuzes shall contain:

a. Detonators from one interfix lot number from not more than three lots from one manufacturer.
b. Primers from one interfix lot number from not more than three lots from one manufacturer.
c. Metal parts from one interfix lot number from one manufacturer.
d. Safety and arming devices from not more than two inspection lots.

4.3.1.1 Process specification for plastic parts. - Prior to the start of normal production, the contractor shall obtain approval from the procuring activity for the process specification of

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plastic parts submitted with the first article sample. This shall specify the time, temperature and pressure ranges which shall be maintained during production of the molded parts. At unscheduled intervals during the operation of the approved process, the Government inspector shall observe the process to ascertain that the approved limits are being maintained. No deviation from these limits without approval from the procuring activity shall be allowed. In the event of an approved deviation the contractor shall submit a first article sample.

4.3,2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Washer, Support (see dwg. 9269699 covering a detail of dwg. 9268173).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Outside diameter.	Gage	01001
102.	Inside diameter	Gage	01002
103.	Thickness	Gage	01003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	01004

4.3.2.2 Washer, Anti-setback (see dwg. 9269698 covering a detail of dwg. 9268179).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined.

Major: None defined.

Minor:		AQL 1.50 percent	
201.	Distance between tabs	Gage	02001
202.	Width of tabs, min	Gage	02002
203.	Thickness	Gage	02003
204.	Evidence of poor workmanship (see 3.5)	Visual	02004

4.3.2.3 Pin, Firing. (see dwg. 9269604 covering a detail of dwg. 926817).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.50 percent		
101.	Large outside diameter	Gage	03001
102.	Outside diameter of striker pin cavity	Gage	03002
103.	Diameter of striker pin cavity . .	Gage	03003
104.	Diameter of nose shaft	Gage	03004
105.	True position of nose shaft to large outside diameter	Gage	03005
106.	True position of outside diameter of striker pin cavity to large outside diameter	Gage	03006
107.	True position of striker pin cavity diameter to large outside diameters	Gage	03007
108.	True position of nose point to large outside diameter	Gage	03008
109.	Length from nose point to rear of firing pin shoulder . .	Gage	03009
110.	Length from aft end to firing pin shoulder	Gage	03010
111.	Depth of striker pin cavity. . .	Gage	03011
112.	Width of large outside diameter (firing pin shoulder)	Gage	03012
113.	Profile of point	Gage	03013
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	03014

4.3.2.4 Mass, Pin, Firing (see dwg. 9269693 covering a detail of dwg. 68179).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 2.50 percent		
101.	Overall length, max	Gage	04001
102.	Length from aft end to angled edge of retaining spring slot, min	Gage	04002
103.	Angle of retaining spring groove	Gage	04003
104.	Outside diameter	Gage	04004
105.	Diameter of firing pin body cavity	Gage	04005
106.	Diameter of firing pin spring cavity, min	Gage	04006
107.	Diameter of firing pin shoulder cavity	Gage	04007
108.	Diameter of washer cavity	Gage	04008
109.	True position of firing pin shoulder cavity to firing pin body cavity	Gage	04009
110.	True position of outside diameter to firing pin body cavity	Gage	04010
111.	True position of washer cavity to firing pin shoulder cavity	Gage	04011
112.	Depth of firing pin spring cavity,	Gage	04012
113.	Depth to firing pin cavity Shoulder	Gage	04013
114.	Depth of washer cavity	Gage	04014
115.	Depth of retaining spring groove	Gage	04015
116.	Width of retaining spring groove	Gage	04016
117.	Width of outside groove	Gage	04017
118.	Depth of outside groove	Gage	04018
119.	Finish of firing pin body cavity	Gage	04019
120.	Finish of outside diameter	Gage	04020
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	04021

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4.3.2.5 Spring, retaining (see dwg. 9269703 covering a detail of dwg. 9268180).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Width	Gage	05001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	05002

4.3.2.6 Sleeve, Pin, Firing (see dwg. 9269695 covering a detail of dwg. 9268180).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.50 percent		
101.	Outside diameter, max	Gage	06001
102.	Inside diameter, min	Gage	06002
103.	Diameter of small retaining spring slot	Gage	06003
104.	Diameter of large retaining spring slot	Gage	06004
105.	Runout of inside diameter to outside diameter	Gage	06005
106.	Runout of small retaining spring slot diameter to outside diameter	Gage	06006
107.	Runout of large retaining spring slot diameter to outside diameter	Gage	06007
108.	Parallelism of edge of small retaining spring slot to end of sleeve	Gage	06008
109.	Overall length, max	Gage	06009
110.	Length from unchamfered end of sleeve to small retaining spring slot	Gage	06010
111.	Overall width of small and large retaining spring slots	Gage	06011
112.	Width of large retaining spring slot	Gage	06012

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113.	Finish of small retaining spring slot	Gage	06013
114.	Finish of inside diameter	Gage	06014
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	06015

4.3.2. Firing Pin and Sleeve Assembly (see dwg. 9268180 covering a detail of dwg. 9271602).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Length of protrusion of firing pin beyond end of sleeve	Gage	07001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	07002

4.3.2.8 Spring, firing pin. (see dwg. 9269700 covering a detail of dwg. 9271602).

Categories	Defect	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Outside diameter	Gage	08001
102.	Solid height	Gage	08002
103.	Load at minimum height	Gage	08003
104.	Load at maximum height	Gage	08004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	08005

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4.3.2.9 Support, Timer (see dwg. 9269697 covering a detail of dwg. 9268199.

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major :	AQL 1.50 percent		
101.	Outside diameter, max	Gage	09001
102.	Diameter of cushion groove, min .	Gage	09002
103.	Width of cushion groove, min	Gage	09003
104.	Diameter of countersunk hole min.	Gage	09004
105.	Diameter of center hole	Gage	09005
106.	Distance between holes	Gage	09006
107.	Radius of rotor cut out	Gage	09007
108.	Radius of roller lock cut out .	Gage	09008
109.	Radius of outer plate cut out .	Gage	09009
110.	Depth of rotor cut out	Gage	09010
111.	Depth of roller lock cut out . .	Gage	09011
112.	Depth of outer plate cut out . .	Gage	09012
113.	Width of rotor cut out	Gage	09013
114.	Width of roller lock cut out . .	Gage	09014
115.	Length from edge of timer to rotor cutout	Gage	09015
116.	Width of rotor housing arm slot .	Gage	09016
117.	Distance from edge of timer to outer plate cut out	Gage	09017
118.	Overall thickness.	Gage	09018
Minor:	AQL 1.50 percent		
201.	Depth of cushion groove	Gage	09019
202.	Distance from outer plate cut out to post recess	Gage	09020
203.	Length of tab on post	Gage	09021
204.	Evidence of poor workmanship (see 3.5)	Visual	09022

4.3.2.10 Timer Support Assembly (see dwg. 9268199 covering a detail of dwg. 9271602).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			

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Major:		AQL 0.40 percent		
101.	Width of timer support and cushion		Gage	10001
102.	Cushion firmly cemented in place		Visual	10002
Minor:		AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)		Visual	10003

4.3.2.11 Housing, Timer (see dwg. 9269692 covering a detail of dwg. 9271602.

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.50 percent		
101.	Outside diameter	Gage	11001
102.	Inside diameter	Gage	11002
103.	True position of inside diameter to outside diameter	Gage	11003
104.	Diameter of center hole	Gage	11004
105.	True position of center hole to outside diameter	Gage	11005
106.	Diameter of node	Gage	11006
107.	True position of node to center hole	Gage	11007
108.	Height of node	Gage	11008
109.	Wall thickness	Gage	11009
110.	Length of inside wall to end of tab	Gage	11010
111.	Length of tab	Gage	11011
112.	Width of tab	Gage	11012
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	11013

4.3.2.12 Weight, setback (see dwg. 8883728 for a detail of dwg. 9271602).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Diameter of stud holes, max	Gage	12001
102.	Width of rotor housing arm slot, max	Gage	12002

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103.	Location of stud holes and rotor housing arm slot	Gage	12003
104.	Chamfer at base missing	Visual	12004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	12005

4.3.2.13 Rotor Housing Assembly (see dwg. 8883719 for a detail of dwg. 9271602).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Arm width (nearest studs), min.	Gage	13001
102.	Location of studs and nearest arm	Gage	13002
Minor:	AQL 1.00 percent		
201.	Surface finish on arm incorrect (nearest studs)	Visual	13003
202.	Evidence of poor workmanship (see 3.5)	Visual	13004

4.3.2.14 Device, Safety and Arming (see dwg. 9215619-1 covering a detail of dwg. 9271602).

Categories	Defects	Method of Inspection	Code No.
Critical:	100 percent		
1.	Distance between annular gear and No. 2 gear, max.	Gage	14001
Major:	AQL 0.40 percent		
101.	Alignment of detonator cavity with the outside diameter across the rotor housing posts . .	Gage	14002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	14003

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4.3.2.15 Device, Safety and Arming (see dwg. 9225617-1 covering a detail of dwg. 9271602).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.5 percent		
101.	Height from forward face of base to top of rotor, max. . . .	Gage	15001
102.	Distance from rotor flat to edge of base, min.	Gage	15002
103.	Primer inverted (the red disc should be observed through top end).	Visual	15003
104.	Primer not staked securely in place	Visual	15004
105.	Primer perforated or deformed by stake	Visual	15005
106.	Detonator inverted (the yellow disc should be observed through aft end	Visual	15006
107.	Detonator perforated or deformed by stake	Visual	15007
108.	Foreign matter present	Visual	15008
109.	Lubricant missing	Visual	15009
110.	Diameter across posts of rotor housing, min.	Gage	15010
111.	protective finish missing or incomplete (all visible plated parts excluding wear and staked areas	Visual	15011
112.	Markings missing or incomplete .	Visual	15012
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	

4.3.2.16 Housing, Booster (see dwg. 8883682 covering a detail of dwg. 9271602).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 1.5 percent		
101.	Total length	Gage	16001
102.	Location of counterbore from bottom of booster cavity	Gage	16002
103.	Diameter of booster cavity . .	Gage	16003
104.	Diameter of lead cavity	Gage	16004
105.	Depth of lead cavity	Gage	16005
106.	Height of lead	Gage	16006
107.	Thickness at end of lead cavity	Gage	16007
108.	Width of O-ring groove	Gage	16008
109.	Diameter of O-ring groove . . .	Gage	16009
110.	Concentricity of thread pitch diameter with booster cavity	Gage	16010
111.	Concentricity of lead cavity with booster cavity.	Gage	16011
112.	Major diameter of threads. . . .	Gage	16012
113.	Concentricity of O-ring groove with pitch diameter	Gage	16013
114.	Finish of O-ring groove	Gage	16014
Minor:	AQL 2.5 percent		
201.	Width across wrench slots. . . .	Gage	16015
202.	Symmetry of wrench slots	Gage	16016
203.	Inside diameter at crimp section	Gage	16017
204.	Wall thickness at crimp section	Gage	16018
205.	Protective finish missing or incomplete	Visual	16019
206.	Evidence of poor workmanship (see 3.5)	Visual	16020

4.3.2.17 Body (see dwg. 9269691 covering a detail of dwg. 9268177).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined,		
Major:	AQL 2.50 percent		
101.	True position of striker pin hole with S&A cavity	Gage	17001
102.	True position of spring cavity with S&A cavity	Gage	17002

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103.	True position of firing pin mass cavity with S&A cavity.	Gage	17003
104.	True position of booster thread with S&A cavity	Gage	17004
105.	Diameter of S&A cavity, min. . . .	Gage	17005
106.	Diameter of striker pin hole, min	Gage	17006
107.	Diameter of spring cavity, min. .	Gage	17007
108.	Diameter of firing pin mass cavity, min	Gage	17008
109.	Minor diameter of booster thread cavity, min.	Gage	17009
110.	True position of warhead mounting threads to booster threads . .	Gage	17010
111.	True position of nose cap mounting surface to outside diameter	Gage	17011
112.	True position of aft diameter to booster threads	Gage	17012
113.	Major diameter of warhead mounting thread, min	Gage	17013
114.	Diameter of nose cap mounting surface	Gage	17014
115.	Aft diameter	Gage	17015
116.	Length from aft face to warhead mounting shoulder	Gage	17016
117.	Length from aft face to firing pin mass cavity	Gage	17017
118.	Length of firing pin mass cavity	Gage	17018
119.	Length of spring cavity	Gage	17019
120.	Length from nose to aft end of firing pin mass cavity	Gage	17020
121.	Distance from nose cap mounting shoulder to center line of stud hole	Gage	17021
122.	Length from firing pin mass cavity shoulder to center line of stud holes	Gage	17022
123.	Diameter of stud hole	Gage	17023
124.	Depth of stud hole	Gage	17024
125.	Length from nose to O-ring groove	Gage	17025
126.	Width of O-ring groove	Gage	17026
127.	Finish of O-ring groove	Gage	17027
128.	Finish of firing pin hole	Gage	17028

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Minor:		AQL 4.00 percent		
201.	True position of wrench slots to outside diameter		Gage	17029
202.	Width of wrench slots		Gage	17030
203.	Finish improper		Gage	17031
204.	Diameter of machine service groove		Gage	17032
205.	Width of machine service groove		Gage	17033
206.	Runout of machine service groove with warhead mounting thread		Gage	17034
207.	Length from warhead mounting shoulder to machine service groove		Gage	17035
208.	Evidence of poor workmanship (see 3.5)		Visual	17036

4.3.2.18 Body Assembly (see dwg. 9268177).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Gap between body and stud head.	Gage	
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	18002

4.3.2.19 Pin Striker (see dwg. 9269696 covering a detail of dwg. 9271602).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Length, min.	Gage	19001
102.	Diameter, min	Gage	19002
103.	Finish	Gage	19003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	19004

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4.3.2.20 Cap, Nose (see dwg. 9269701 covering a detail of dwg. 9268178).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.00 percent		
101.	Inside diameter	Gage	20001
102.	Distance from bottom of striker plate cavity to end of cap . .	Gage	20002
103.	Distance from bottom of striker plate cavity to stud slot . . .	Gage	20003
104.	Perpendicularity of aft end of cap to inside diameter	Gage	20004
105.	Width of slot, four places	Gage	20005
106.	True position of slots to inside diameter	Gage	20006
107.	Radius at end of slot	Gage	20007
108.	True position of radius at end of slot to inside diameter. . .	Gage	20008
109.	Distance from centerline of slot to radius at end of slot	Gage	20009
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	20010

4.3.2.21 Nose Cap Assembly (see dwg. 9268178 covering a detail of dwg. 9271602).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major :	AQL 0.40 percent		
101.	Length from striker plate face to aft end of nose cap	Gage	21001
102.	Marking present and legible. . .	Visual	21002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	21003

4.3.2.22 Fuze, Rocket, M423, Modified (Bunker) Less Booster and Lead (see dwg. 9271602).

Critical: None defined.

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Major:	AQL 1.00 percent		
101.	Striker pin missing	Visual	22001
102.	O-rings missing	Visual	22002
103.	Marking missing, incomplete. . .	Visual	22003
104.	Booster housing improperly torque	Gage	22004
105.	Length from aft face to end of booster housing, max	Gage	22005
106.	Presence of retaining clamp. . .	Visual	22006
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	22007

4.3.3 Testing

4.3.3.1 Spring embrittlement (see 3.2.1) - Major defects; detent spring - Code No. 23001 - set-back weight spring - Code No. 24001 - firing pin spring - Code No. 25001 - nose cap spring - Code No. 26001. - A sample of 10 springs shall be selected from each heat treatment batch. The sample shall be tested as specified in 4.4.1 using equipment in accordance with 4.3.4. The heat treatment batch shall be rejected if one or more springs fail to comply with the specified requirement. Springs tested as specified shall be discarded as scrap.

4.3.3.2 Setback weight spring set (see 3.2.2) - Major defect - Code No. 27001. - A sample of ten (10)setback weight springs shall be selected from each heat treatment lot and shall be tested as specified in 4.4.2 using equipment in accordance with 4.3.4. If one or more springs fail to comply with the specified requirement they shall be classed defective and the heat treatment lot shall be rejected. Springs tested as specified shall be discarded as scrap.

4.3.3.2.1 Retaining spring set - Major defect - Code No. 28001. - A sample of ten retaining springs shall be selected from each lot and shall be tested as specified in 4.4.3 using equipment in accordance with 4.3.4. If one or more springs fail to comply with the specified requirement they shall be classed defective and the lot shall be rejected. Springs tested as specified shall be discarded as scrap.

4.3.3.3 Mechanical Tests

4.3.3.3.1 Critical defects. - These tests shall be performed 100 percent. Assemblies falling to comply with the specified

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requirements shall be classed defective and removed from the lot. The tests shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

a. Load test of No. 1 pinion (see dwg. 8883703) Code No. 29001

b. Load test of No. 2 pinion (see dwg. 8883700) Code No. 30001

c. Load test of escapement pinion (see dwg. 8883710) Code No. 31001

d. Load test of lever shaft (see dwg. 8883706) Code No. 32001

e. Load test of lever pallet (see dwg. 8883706) Code No. 33001

f. Load test of annular gear (see dwg. 8883744) Code No. 34001

4.3.3.3.2 Critical defects. - The sampling plan for these tests shall be in accordance with MIL-STD-105 using inspection level II and an AQL of 0.040 percent for each defect. Continuous sampling may be used in accordance with MIL-STD-1235 using inspection level II, CSP-I and an AQL of 0.015 percent for each defect. The tests shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

a. Load test of lock roller pin (see dwg. 8883744) Code No. 35001

b. Torque test of lever shaft (see dwg. 8883706) Code No. 36001

c. Load test of bridge rivet (see dwg. 8883693) Code No. 37001

4.3.3.3.3 Major defects. - The sampling plan for these tests shall be in accordance with MIL-STD-105, using inspection level II and an AQL of 0.40 percent for each defect. The tests shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

a. Load test of spring stud (see dwg. 8883719) Code No. 38001.

b. Load test of pillars (see dwg. 8883692) Code No. 39001.

c. Torque test of rotor stop stud (eccentric pin) (see dwg. 9215619) Code No. 40001.

4.3.3.4 Firing Pin Assembly Push Test - (see dwg. 9268179) Major Defect - Code No. 07003. - This test shall be performed 100 percent as specified in paragraph 4.4.5. Any assembly failing to meet the specified requirement shall be classed defective and removed from the lot. The test shall be performed with equipment in accordance with 4.3.4.

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4.3.3.5 Fuze Body Assembly Seal Leakage (see dwg. 9254602) - Major Defects - Code No. 22008. - The seal leakage test shall be performed on a sample selected in accordance with MIL-STD-105 using Inspection Level II on an AQL of 0.40 percent. This test shall be performed as specified in 4.4.17 using equipment in accordance with 4.3.4.

4.3.3.6 Testing of Anodic Coating. - The testing of anodic coating shall be in accordance with MIL-A-8625 except that panels shall not be used and Paragraph 4.3, "Lot" and 3.10, "Detail Requirements" shall be deleted and the following substituted:

Corrosion Resistance of Anodic Coating - Major Defect - Code No. 20011. - The samples of nose caps for corrosion resistance shall be selected in accordance with MIL-STD-105 using Inspection Level S-2. Lot size for each nose cap shall be of not more than 40 hours of anodic production. If one or more of the samples fail the requirements, the lot shall be rejected. The samples shall be tested as specified in accordance with 4.4.6 using equipment in accordance with 4.3.4. Components tested as specified shall be discarded as scrap.

4.3.3.7 Detonator Push Test (see dwg. 9215617) - Major Defect - Code No. 15014. The sampling plan for this test shall be in accordance with MIL-STD-105 using Inspection Level II and an AQL of 0.40 percent. This test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4.

4.3.3.8 Check Test for Deterioration of Detonators. - If the total time between original acceptance of any detonator lot and the assembly of that lot into the fuzes exceeds two years at any time since previous tests, the detonator lot shall be subjected to, and must satisfactorily pass, the functioning and output tests specified in MIL-D-18665 immediately before the detonator lot Is assembled into the fuzes. This test shall be performed by the contractor on detonators selected by the Government Inspector at the facility assembling the detonators into the rotors. The test shall be conducted at Government expense without cost to the contractor who loaded the detonators or the contractor who assembled the detonators into the fuzes. The test shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the detonator lot was under the jurisdiction of either contractor.

4.3.3.9 Check Test for Deterioration of Primers. If the total time between original acceptance of any primer lot and the assembly of that lot into the fuzes exceeds two years at any time

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since previous tests, the primer lot shall be subjected to, and must satisfactorily pass the functioning and output tests specified in AEI-93 immediately before the primer lot is assembled into the fuzes. This test shall be performed by the contractor on primers selected by the Government Inspector at the facility assembling the primers into the rotors. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or the contractor who assembled the primers into the fuzes. The test shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer was under the jurisdiction of either contractor.

4.3.3.10 Arming - Code No. 15015; Detent functioning - (see 3.3.1) Code No. 15016 - Major defects. - These tests shall be performed 100 percent on the loaded S&A. S&A's failing to comply with the specified requirements shall be classed defective and removed from the lot. The tests shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.3.11 Setback weight return capability (see 3.3.2) - Critical defect - Code No. 15017.- This test shall be performed 100 percent on the loaded safety and arming devices. Safety and arming devices failing to comply with the specified requirements shall be classed defective and removed from the lot. This test shall be performed as specified in 4.4.9 using equipment in accordance with 4.3.4.

Defect standards with setback weights that bind when depressed and remain in the depressed condition shall be introduced at various times to verify test performance. If inspection personnel fail to identify the standard as defective, all material accepted subsequent to the last acceptable verification check shall be reinspected.

4.3.3.12 Rotor reset capability (see 3.3.3) - Critical defect - Code No. 15018 This test shall be performed 100 percent on the loaded S&A. S&A’s failing to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.1.0 using equipment in accordance with 4.3.4.

4.3.3.13 Arming Time - Major defect - Code No. 15019.- This test shall be performed 100 percent on the loaded S&A. S&A’s failing to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified 4.4.11 using equipment in accordance with 4.3.4.

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4.3.3.14 Non-Arming. - Critical defect - Code No. 15020. - This test shall be performed to 100 percent just prior to insertion of S&A into the fuze body. S&A's failing to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.12 using equipment in accordance with 4.3.4.

4.3.3.15 S&A position check - Critical defect - Code No. 22010. - The S&A position, after insertion into the fuze body but prior to assembly of the booster housing, shall be inspected 100 percent. Any armed fuze shall be considered defective and removed from the lot and the assembly operation shall be halted until operation is corrected to Government satisfaction. The inspection shall be performed to the requirements of 3.3.6 using procedures in accordance with 4.1.2 and equipment in accordance with 4.3.4. Defect standards with semi-armed and fully armed conditions shall be introduced at various times to verify test performance (see 4.1.2.1). The Design Agency approved hanling procedure to preclude untested S&A's from being introduced into the fuze prior to the booster housing being installed shall also be assessed when the standards are introduced. If an improper procedure is being used as determined by the Government, all fuzes accepted subsequent to the last acceptable handling procedure assessment shall be screened as determined by the Government.

4.3,3,16 Jolt - Code No. 22011; Jumble - Code No. 22012 - Critical defects - A sample of twelve (12) fuze assemblies shall be selected from each lot and be subjected to the test as specified in 4.4.15 using equipment in accordance with 4.3.4. If one or more fuze assemblies fail to comply with the specified requirements, the lot shall be rejected.

4.3.3.17 Forty-foot drop test - Critical defect - Code No. 22013. - A sample of four (4) fuze assemblies shall be selected as part of the First Article Sample only. The fuze assemblies shall be subjected to the test as specified in 4.4.16 using equipment in accordance with 4.3.4.

4.3.3.18 Fuze Assembly Operational Tests (see 3.4.5).- A random sample of forty-five (45) fuze assemblies shall be selected for the Five Foot Drop Test as required in 4.3.3.18.1. A random sample of 315 fuze assemblies, in accordance with MIL-STD-105, sample size M, shall be selected from each lot as the total combined sample for the Non-Environment Test and the Functional Test. Operability of the fuze assemblies in the Non-Environment Test and the Functional Test shall be indicated by subjection to the tests and acceptance or rejection criteria in 4.3.3.18.4. Testing for operability under 4.3.3.18.1 and 4.3.3.18.4 shall be us specified in 4.4.13 and 4.4.14, the firing test shall be performed as specified in 4.4.18 using equipment in accordance with 4.3.4.

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4.3.3.18.1 The Five Foot Drop Test, as specified in MIL-STD-331, shall be performed with the following conditions:

a. A sample of 45 fuzes selected in accordance with 4.3.3.18 shall be subjected to the following test.

b. The fuze assembly will be assembled to a MK1 warhead (The Government shall supply the warhead for the contractors use) , Filler E loaded, and with impact orientation of the warhead fuze assemblies for the drop test as follows:

(1) Major axis horizontal

(2) Major axis 45 degrees from vertical base down

(3) Base down

Fifteen fuzes each must be dropped in the above (1,2,3) orientations. No fuze shall be subjected to more than one drop. The orientation of the safety and arming device (specifically the g-weight) of the fuze, shall be random.

c. The Five Foot Drop Test, as specified in MIL-STD-331, shall be performed until three (3) consecutive lots (a minimum lot size of one week's production) have complied with the acceptance criteria specified. After satisfying the above requirement one (1) lot (a minimum lot size of one week's production) per month shall be selected by the Government and subjected to and pass the Five Foot Drop Test. Should the one lot (per month) fail the acceptance criteria, the contractor shall be required to return to the three (3) consecutive lot procedures above for process qualification.

d. The fuzes shall be subjected to the tests and acceptance or rejection criteria as specified in Table I.

Table I

Sample Size 45

Test	Accept	Reject	Defect	Code No.
Non-Arm	0	1	Critical	22014
Arming Time				
Under Min	0	1	Critical	22015
Over Max	2	3	Major	22016
Fail to Arm	1	2	Major	22017
Firing	1	2	Major	22018

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After the arming time test, twenty (20) fuzes shall be fired in the point detonating mode, Major defect - Code No. 22018 and twenty-five (25) fuzes shall be fired in the bunker rode, Major defect - Code No. 22019. The results of these two firings shall be added and subjected to the acceptance/rejection criteria of Table I. If the sample fails the above acceptance criteria the lot shall be rejected.

4.3.3.18.2 Non-Environment Test.- A sample of 219 fuze assemblies shall be subjected to the tests and acceptance or rejection criteria as specified in 4.3.3.18.4. This sample may be selected prior to application of the sealing compound to the booster housing threads and may be assembled without sealing compound.

4.3.3.18.3 Functional Test.- 96 fuzes shall be subjected to Aircraft Vibration, Method 4, Curve Z as specified in MIL-STD-810. Vibration conditioning at resonance is not required. After vibration the fuzes shall be subjected to the Seal Leakage Test, 4.3.3.5, and the lot shall be rejected if four (4) or more defects occur. Following seal leakage testing, the fuzes shall be subjected to the tests and acceptance or rejection criteria specified in 4.3.3.18.4.

4.3.3.18.4 Acceptance Criteria. - The sample fuzes from the Non-Environment Test and the Functional Test shall be combined into one sample and subjected to the Non-Arm and Arming Time Tests. If one or more fuzes fail to comply with the Non-Arm requirement - Critical defect - Code No. 22014 - or the Minimum Arming Time requirements - Critical defect - Code No. 22015 - the lot shall be rejected. The Arming Time defects classified as Major defects shall be accepted in accordance with Table II.

TABLE II

Sample Size 315

DEFECT	CODE NO.	POINTS	ACCEPT	REJECT
Arming Time Over Max.	22016	1	5 POINTS	6 POINTS
Failure To Arm	22017	3		

Acceptable fuzes from the 96 sample Functional Test shall be subjected to the firing tests - Major defects - Code No's. 22018 and 22019 - after the Arming Time Test. Forty eight (48) fuzes shall be fired in each of the two modes; point detonating and bunker. The results of both tests shall be combined and if two (2) or more samples fail to fire the lot shall be rejected. If all tests are acceptable the sample fuzes from the Non-Environment Test may be returned to the lot after disassembly, resetting the S&A into the safe position and reassembly with sealing compound.

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4.3.3.18.5 Definitions

a. Times under .63 seconds for the modified M423 are classed as defect Code No. 22015.

b. Times .83 seconds through 1.20 seconds for the modified M423 are classed as defect Code No. 22016.

c. Times over 1.20 and Failures to Arm are classed as defect Code No. 22017.

d. Identification of defects under Code Nos. 22014, 22015, 22016, 22017 will depend on both centrifuge testing and radiographic examination.

e. Any fuze classified defective as per defect 22014, 22015, 22016, 22017 shall be removed and not subjected to firing (see 6.6).

f. All fuzes not classified defective as per defect 22014, 22015, 22016, 22017 shall be radiographically inspected prior to firing.

g. NOTE: If the test equipment fails to produce an arming readout (i.e, recording paper skewes, audio pick-up or amplifier inoperative) during the arming time test as verified by the Government, the safety and arming device shall be removed from the fuze body, and if found to be armed or partially armed the safety and arming device shall be reset into the unarmed position, reassembled into the same fuze body and retimed once as a fuze. This test shall be performed to the timing criteria of this paragraph. The only handling action performed on the safety and arming device shall be the reset of the device just prior to the arming time test. The timing record produced by the fuze on the approved fuze timing equipment shall be considered the timing record of the fuze under the applicable defect classification (22015, 22016, and 22017). Safety and arming devices removed from the fuze body and found to be unarmed shall be classified defective per defect 22017.

h. Fuze arming times, if recorded to the thousandth of a second, shall be “rounded off” to the nearest hundredth of a second. Example: a. 0.625 shall be 0.62
0.626 shall be 0.63
b. 0.825 shall be 0.82
0.826 shall be 0.83

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4.3.4 Inspection Equipment. - Equipment Tabulation Number ET 9271602 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock , ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Spring embrittlement. - The spring shall be stretched along its longitudinal axis, in the case of a coil spring, or bent in the case of a flat spring, until the spring no longer returns to its original position. When the spring no longer returns to its original length or profile, it shall be considered distorted permanently. Any spring that breaks prior to this point shall be classed defective.

4.4.2 Setback weight spring set. - The springs shall be measured at their original free length, assembled in a restraining fixture in accordance with 4.3.4 and compressed to a normal height of .750 inches, placed in an oven and subjected to a temperature of not more than 180 degrees Fahrenheit (F.) or less than 175 degrees F. for 24 hours. The springs shall be removed from the oven and be cooled to ambient temperature. After cooling the springs for 24 hours, min. the springs shall be measured and the measurement shall be compared to the original free length to ascertain compliance with the specified requirement.

4.4.3 Retaining spring set. - The springs shall be measured at their original free height, assembled in the restraining fixture in accordance with 4.3.4 and left in the retainer for 24 hours. After 24 hours the springs shall be removed from the fixture and measured and the measurement shall be compared to the original free height to ascertain compliance with the specified requirement.

4.4.4 S&A Mechanical Tests -

4.4.4.1 Load test of No. 1 gear and pinion assembly an axial load perpendicular to the No. 1 gear and in such a direction as to test the assembly of the No. 1 gear and pinion. The resistive force shall act against the No. 1 gear. Upon removal of the resistive force, the maximum permissible gap shall be in-

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spected for compliance with the drawing requirement.

4.4.4.2 Load test of No. 2 gear and pinion assembly. - Apply an axial load perpendicular to the No. 2 gear and in such a direction as to test the assembly of the No. 2 gear and pinion. The resistive force shall act against the No. 2 gear. Upon removal of the resistive force, the maximum permissible gap shall be inspected for compliance with the drawing requirement.

4.4.4.3 Load test of escapement gear and pinion assembly. - Apply an axial load perpendicular to the escapement gear and in such a direction as to test the assembly of the escapement gear and pinion. The resistive force shall act against the escapement gear. Upon removal of the resistive force, the maximum permissible gap shall be inspected for compliance with the drawing requirement.

4.4.4.4 Load test of lever shaft. - Apply an axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and lever shaft. The resistive force shall act against the lever. Upon removal of the resistive force, the maximum permissible gap shall be inspected for compliance with the drawing requirement.

4.4.4.5 Load test of lever palette. - Apply an axial load perpendicular to the lever and in such a direction as to test the assembly of the lever and the lever pallet. The resistive force shall act against the lever. Upon removal of the resistive force, the height of the lever pallet above the lever shall be inspected for compliance with the drawing requirement.

4.4.4.6 Load test of annular gear. - Apply a load perpendicular to the rotor and in such a direction as to test the assembly of the annular gear studs to the annular gear and the rotor. The resistive force shall act against the rotor. Upon removal of the resistive force, the distance from the face of the rotor to the outside face of the annular gear shall be inspected for compliance with the drawing requirement.

4.4.4.7 Load test of lock roller pin. - Apply an axial load perpendicular to the rotor and in such a direction as to test the strength of the assembly of the lock roller pin to the rotor. The resistive force shall act against the rotor.

4.4.4.8 Torque Test of Lever Shaft. - A Torque shall be applied between the lever and the lever shaft.

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4.4.4.9 Load test of bridge rivet. - Apply the required axial load perpendicular to the inner plate and into the bridge rivet from the side opposite the bridge. The resistive force shall act against the inner plate.

4.4.4.10 Load test of spring stud. - Apply an axial load perpendicular to the rotor housing and in such a direction as to test the assembly of the spring stud and the rotor housing. The resistive force shall act against the rotor housing.

4.4.4.11 Load test of pillars. - Apply the required axial load perpendicular to the inner plate and in such a direction as to test the assembly of the pillars to the inner plate and outer plate. The resistive force shall act against the inner plate.

4.4.4.12 Torque test of rotor stop stud (eccentric pin). - The required torque shall be applied between the rotor stop stud and the gear assembly. The resistive force shall act on the gear assembly.

4.4.5 Firing pin assembly push test. - The firing pin assembly shall be assembled to the test equipment in accordance with paragraph 4.3.4. After assembly, the required force shall be applied at the required time rate to check the crimped area of the assembly in meeting the requirements specified on the drawing.

4.4.6 Corrosion resistance of anodic coating. - The testing of the sample shall be in accordance with the procedures specified in MIL-A-8625 except that the salt spray test shall be conducted for 168 hours. The significant area for examination shall be the outside surface of the specimens. The acceptance criteria shall be:

a. The specimens shall show no more than 5 spots or pits, none larger than 1/32 inch in diameter, except in those areas within 1/16 inch from the edges, corner, identification markings, and holding areas during processing.

b. Powdery areas or loose films in excess of 10 percent of the examined surface shall be cause for rejection.

4.4.7 Load test of detonator. - Apply the required axial load perpendicular to the rotor and in such a direction as to test the strength of the staking of the detonator in the rotor.

WARNING: This test may be dangerous and should be performed behind a suitable barricade.

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4.4.8 Arming and detent functioning. - The S&A in the safe position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the safety and arming device to 20.75 g's maximum for a minimum of 3 plus 1, minus 0 seconds in a direction which simulates rocket launch. At the completion of acceleration, the equipment shall be halted and visual observations shall be made for the locking position of the detent. Finger tip pressure shall be applied to the rotor to rotate in the safe direction to ascertain positive arming. At the completion of the test the safety and arming device shall be returned to the safe position.
WARNING: This test may be dangerous and should be performed behind a suitable barricade.

4.4.9 Setback weight return capability.- The loaded safety and arming device shall be assembled to the test fixture in accordance with 4.3.4. A force of 20 lbs shall be applied equally to the two top surfaces of the setback weight toward the base of the rotor housing, and then released. Observation shall be made for setback weight return to the original fully safe position.

4.4.10 Rotor reset capability. - The loaded S&A shall be assembled to the test fixture in accordance with 4.3.4. The rotor shall be rotated to a position at least 26 degrees but not more than 35 degrees from the fully safe position and released. Observation shall be made for rotor return to the fully safe position.

4.4.11 S&A Arming Time. - The loaded S&A with the g-weight in the downward position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the safety and arming device in a clockwise direction to 27.00 g's plus or minus 0.25 g's and the arming time shall be recorded to the nearest hundredth of a second using a calibrated timer.

4.4.12 S&A Non-Arming. - The loaded S&A with the g-weight in the downward position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the safety and arming device in a clockwise direction to the 11.25 g's minimum for a minimum of 3 seconds. Observation shall be made for rotor in the safe position.

4.4.13 Fuze Arming Time. The fuze assembly with the g'weight in the downward position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the fuze in a clockwise direction to 27.00 g's plus or minus 0.25 g's and the arming time shall be recorded to the nearest hundredth of a second using a swing fixture and acoustic pick-up feeding the arming cycle noise signal to a design agency approved recorder.

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4.4.14 Fuze Non-arming. - The fuze assembly with the g'weight in the downward position shall be assembled to the test equipment in accordance with 4.3.4. The equipment shall accelerate the fuze assembly in a clockwise direction to 11.50 g’s plus or minus 0.25 g’s for a minimum of 3 seconds. X-ray examination shall be made for rotor in the safe positions.

4.4.15 Jolt and Jumble. - The fuze assembly shall be subjected to the Jolt test specified in Standard MIL-STD-331 with the exception that the fuze shall not be disassembled and inspected after the Jolt test. After completion of the Jolt test the same fuze shall be subjected to and withstand the Jumble test specified in MIL-STD-331.

4.4.16 Forty Foot Drop. - The fuze assembly with warhead assembly shall be subjected to the Forty Foot Drop test as specified in MIL-STD-331 with the exception that the complete round shall be dropped base down only in a complete pack. The pack is as specified on dwg.s 8886810 and 8886811.

4.4.17 Seal Leakage test.- The fuze assembly shall be checked for leakage in to the S&A cavity. Fuze assemblies in accordance with dwg. 9271602 shall be tested for the leakage rate past the “O” ring in the booster cavity with the application of 5.0 plus or minus 0.25 psig to the base of the fuze. The integrity of the seals shall be verified by measuring the pressure drop per unit time. A leak into the S&A cavity in excess of 0.00072 psi in 10 sec. shall indicate excessive leakage and reject the fuze. The test equipment calibration and equipment sensitivity shall be in accordance with 4.3.4.

4.4.18.1 Point detonating mode. The armed fuze shall be installed in the test fixture in accordance with 4.3.4 and a mass of 50 to 51 pounds shall be dropped from a height of 5.0 to 5.5 feet onto the nose of the fuze. The weight shall be dropped only once.

4.4.18.2 Bunker mode. - The armed fuze shall be installed in the drop fixture in accordance with 4.3.4 and then dropped from a height of three feet. The fuze shall be dropped in the nose-down configuration such that the fuze nose strikes a steel plate. The fuze shall be dropped only once.

5. PREPARATION FOR DELIVERY

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5.1 Packing and marking for the Fuze, Rocket, M423, Modified (Bunker) less Booster and Lead shall be as specified on drawings 9275050 and 9275051.

6. NOTES

6.1 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-166.

6.2 Inspection code numbers. - The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as “Contractor Design” in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Design shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.

MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information

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necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five digit Code-Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-F. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal

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completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of First Article Samples. - Instructions as to the location for the evaluation of the first article sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquires should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-F, Dover, New Jersey 07801.

6.5 Government furnished parts.

6.5.1 Packing and packaging.

6.5.2 M151 Warheads. - The procuring agency will provide inert loaded F151 Warheads for the Forty Foot Drop Test.

6.5.3 MK1 Warheads.- The procuring agency will provide inert loaded MK1 warheads for the 5 Foot Drop Test.

6.6 Defect reporting. - From each lot produced, the contractor shall submit two copies of the Metal Parts Lot Acceptance Test Data, to Picatinny Arsenal, ATTN: SMUPA-QA-A-F and SMUPA-AD-F-E-4; within three (3) days after completion of acceptance testing of the lot. If fuzes and/or tests in addition to acceptance requirements were utilized to evaluate the lot, the results of these tests will also be submitted three (3) days after the additional tests are completed. The defective fuzes shall be saved pending review of the results.

6.7 Intended use. - It is intended that the item of this specification will be used in the 2.75 Inch rocket.

6.8 The following drawing numbers shall be preceded by the letter T:

Dwg. No.	Dwg. No.	Dwg. No.
9271602	9269695	9269691
9269699	9268180	9268177
9269698	9269700	926%96
9269694	9269697	9269701
9269693	9268199	9268178
9269703	9269692	9271602
		9268179

Custodian:
Army-MU

Preparing Activity:
Army-MU

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 64 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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